Digital Logic System

Tutoriol 1

1. Convert the following
   i) $(225.225)_{10}$ to $(----)_2$, $(----)_8$, $(----)_{16}$
   ii) $(1001001.011)_2$ to $(----)_{10}$, $(----)_8$, $(----)_{16}$
   iii) $(623.77)_8$ to $(----)_2$, $(----)_{10}$, $(----)_{16}$
   iv) $(2AC5.D)_{16}$ to $(----)_2$, $(----)_{10}$, $(----)_8$
   v) $(250.5)_{10}$ to base-3,4 and 7.
2. Obtain 1's and 2's complement of:
   1010101,0111000, 0000001,10101110
3. Obtain 9's and 10's complement of:
   13579, 09900, 90090, 10000
4. Subtract the following decimal numbers using 9's and 10's complement:
   i) 5250-321 iiI) 753-864
   ii) 3570-2100 IV) 20-100
5. Subtract the following binary numbers using 1's and 2's complement:
   i) 11010-1101 iii) 10010-10011
   ii) 11010-10000 iv) 100-110000
6. Define analog signal and digital signal. Write the differences between analog and digital system.
7. Define analog and digital systems. Why do you prefer digital systems over its analog counterparts?
8. 'Excess-3 code is also known as self-complementing code'.
9. Define weighted and non-weighted codes with examples.
10. Show your familiarization with Gray code. Write the 4-bit gray code. Also explain why Gray code is also known as reflected code.
11. Perform the following:
    i) $(84.9)_{10}=(----)_{BCD}$, $(----)_{excess\ 3}$
    ii) $(1101)_{Gray}=(----)_2$
    iii) $(110101)_{Gray}=(----)_2$
    iv) $(101111)_2=(----)_{Gray}$
    v) $(63)_8 = (----)_{excess\ 3}$
12. Write short notes on:
    I) Alphanumeric code
    II) Instruction code

Date
Page

1 Convert the following

i (225.225) to $(?)_2$, $(?)_8$ $(?)_{16}$

Soln

$= (225.225)_{10}$ to $(?)_2$

2	225	R
2	112	1
2	56	0
2	28	0
2	14	0
2	7	0
2	3	1
	1	1

$0.225 \times 2 = 0.45$
$0.45 \times 2 = 0.9$
$0.9 \times 2 = 1.8$

$\therefore (225.225)_{10} = (11100001.001)_2$

$= (225.225)_{10}$ to $(?)_8$

8	225	R
8	28	1
	3	4

$0.225 \times 8 = 1.8$
$0.8 \times 8 = 6.4$
$0.4 \times 8 = 3.2$
$0.2 \times 8 = 1.6$

$\therefore (225.225)_{10} = (341.1631)_8$

$(225.225)_{10}$ to $(?)_{16}$

$= 16 \mid 225 \quad R$

$16 \mid 14 \quad E \quad 1$

$0.225 \times 16 = 3.6$

$0.6 \times 16 = 9.6$

$0.6 \times 16 = 9.6$

$0.6 \times 16 = 9.6$

$\therefore (225.225)_{10} = (E1.3999)_{16}$

ii $(1001001.011)_2$ to $(?)_{10}$, $(?)_8$, $(\ )_{16}$

FOR Binary to Decimal

$6\ 5\ 4\ 3\ 2\ 1\ 0\ .\ -1\ -2\ -3$

$1\ 0\ 0\ 1\ 0\ 0\ 1\ .\ 0\ 1\ 1$

$= 1 \times 2^6 + 0 \times 2^5 + 0 \times 2^4 \times 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 0 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3}$

$= 64 + 8 + 1 + \frac{1}{2^2} + \frac{1}{2^3}$

$= 73.375$

$= (73.375)_{10}$

iii $(623.77)_8$ to $(?)_2$ $(?)_{10}$, $(?)_{16}$

Soln.

Octal to Binary

623 . 77

$\therefore (623.77)_8 = (110010011.111111)_2$

6 = 1 1 0
2 = 0 1 0
3 = 0 1 1
7 = 1 1 1
7 = 1 1 1

Octal to Decimal

$6\overset{2}{2}\overset{1}{3}\overset{0}{.}\overset{-1}{7}\overset{-2}{7}$

$= 6\times8^2 + 2\times8^1 + 2\times8^0 + 7\times8^{-1} + 7\times8^{-2}$

$= 384 + 16 + 2 + \frac{7}{8} + \frac{7}{8^2}$

$= (402.984375)_{10}$

Octal to Hexadecimal

623 . 77 = 000110010011 . 11111100

Taking 4 digites because 4 Digits of binary is equivalent to single digit of hexadecimal

8 4 2 1
0 0 0 1 = 1
1 0 0 1 = 9
0 0 1 1 = 3
1 1 1 1 = F
1 1 0 0 = C

$\therefore (623.77)_8 = (193.FC)_{16}$

iv $(7AC5.D)_{16}$ to $(?)_2$, $(?)_{10}$ $(?)_{16}$

Soln

To Binary

7	A	C	5	.	D
0 0 1 0	1 0 1 0	1 1 0 0	0 1 0 1		1 1 0 1

$\therefore (7AC5.D)_{16} = (0010\,1010\,1100\,0101.1101)_2$

8 4 2 1 Taking 4 Digits because

7 = 0 0 1 0

A = 1 0 1 0 4 Digits in Binary is equivalent

C = 1 1 0 0

5 = 0 1 0 1 to 1 Digit is Hexadecimal

D = 1 1 0 1

Date
Page

To Decimal

$$(2^{3}\ A^{2}\ (^{1}\ 5^{0}\ .\ D^{-1})_{16}$$

$$= 2 \times 16^{3} + A \times 16^{2} + C \times 16^{1} + 5 \times 16^{0} + D \times 16^{-1}$$

$$= 2 \times 16^{3} + 10 \times 16^{2} + 12 \times 16^{1} + 5 \times 16^{0} + 13 \times 16^{-1}$$

$$= 8192 + 2560 + 192 + 5 + \frac{13}{16}$$

$$= (10949.8125)_{10}$$

To Octal

2 A C 5 . D

0010 1010 1100 0101 1101

000010101011000101000 . 110100

Taking 3 Digits

$$\therefore (25305.64)_{8}$$

921

000

010

101

011

Date
Page

v (250·5)10 to base 3, 9 and 7

for base 3

~~7 5 0~~

#3	250	R
#3	#83	1
#3	27	2
#3	#9	0
#3	3	0
	#1	0

0·5 x 3 = 1·5
0·5 x 3 = 1·5
0·5 x 3 = 1·5
0·5 x 3 = 1·5

∴ (250·5)10 = (120001·1111)3

for base 4

4	250	R
4	62	2
4	15	2
	3	3

0·5 x 4 = 2

∴ (250·5)10 = (3322·2)4

for base 7

7	250	R
7	35	5
	5	0

0·5 x 7 = 3·5
0·5 x 7 = 3·5
0·5 x 7 = 3·5
0·5 x 7 = 3·5

∴ (250·5)10 = (505·3333)7

Date
Page

2 Obtain 1's and 2's complement of :

a 1010101

Here, $N = 1010101$, $r = 2$, $n = 7$

then

1's complement of $N = N' = (r^n - 1) - N$

$= 1111111 - 1010101$

$= 0101010$

2's complement of $N = N' = r^n - N$

$= 10000000 - 1010101$

$= 0101011$

b 0111000

Here, $N = 0111000$ $r = 2$ $n = 7$

then

1's complement of $N = N' = (r^n - 1) - N$

$= 1111111 - 0111000$

$= 1000111$

2's complement of $N = N' = r^n - N$

$= 10000000 - 0111000$

$= 1001000$

c 0000001

Here, $N = 0000001$ $r = 2$ $n = 7$

then

1's complement of $N = N' = (r^n - 1) - N$

$= 1111111 - 0000001$

$= 1000000 \; 1111110$

Again,

2's complement of $N = N' = r^n - N$

$= 10000000 - 0000001$

$= 1111111$

d 10101110

Here, $N = 10101110$ $r = 2$ $n = 8$

then

1's complement of $N = N' = (r^n - 1) - N$

$= 11111111 - 10101110$

$= 01010001$

Again,

2's complement of $N = N' = r^n - N$

$= 100000000 - 10101110$

$= 01010010$

Date
Page

3 Obtain 9's and 10's complement of

a $(13579)_{10}$

Here,

$N = 13579 \quad r = 10 \quad n = 5$

then

9's complement of $N = N' = (r^n - 1) - N$

$= 99999 - 13579$

$= (86420)_{10}$

Again

10's complement of $N = N' = r^n - N$

$= 100000 - 13579$

$= (86421)_{10}$

b $(09900)_{10}$

Here,

$N = 09900 \quad r = 10 \quad n = 5$

then

9's complement of $N = N' = (r^n - 1) - N$

$= 99999 - 09900$

$= (90099)_{10}$

Again,

10's complement of $N = N' = r^n - N$

$= 100000 - 09900$

$= (90100)_{10}$

Date
Page

c $(90090)_{10}$

Here, $N = 90090$ $r = 10$ $n = 5$

then

9's complement of $N = N' = (r^n - 1) - N$

$= 99999 - 90090$

$= (9909)_{10}$

Again,

10's complement of $N = N' = r^n - N$

$= 100000 - 90090$

$= (9910)_{10}$

d $(10000)_{10}$

Here, $N = 10000$ $r = 10$ $n = 5$

then

9's compliment of $N = N' = (r^n - 1) - N$

$= 99999 - 10000$

$= (89999)_{10}$

Again,

10's compliment of $N = N' = r^n - N$

$= 100000 - 10000$

$= (90000)_{10}$

Date
Page

4 Subtract the following decimal numbers using 9's and 10's complement.

; 5250 - 321

Here, M = 5250, N = 0321

9's complement of N = N' = 999 - 321
= 9678

then,

M + N'

$$\begin{array}{r} M = 5250 \\ N = +9678 \\ \hline 14928 \end{array}$$

$$\begin{array}{r} {}^{1}\\ M = 5250 \\ N = +9678 \\ \hline 14928 \\ \quad\quad +1 \\ \hline 4929 \end{array}$$

∴ Answer = 4929

Again

10's complement of N = N' = 10000 - 321
= 9679

then

M + N'

$$\begin{array}{r} M = 5250 \\ N' = +9679 \\ \hline 14929 \end{array}$$

∴ Answer = 4929

ii 753 - 869

Here

M = 753 N = 869

Now,

9's complement of N = N' = 999 - 869
= 135

then

M + N'

M = 753

N' = + 135

888

No end carry, the result is in 9's complemnt form

∴ exact result = - (9's complemnt of 888)

= - (999 - 888)

= - 111

Again

10's complement of N = N' = 1000 - 869
= 136

then

Date
Page

$M + N'$

$M = 753$

$N = +136$

889

No end carry, the result is in 10's complement form

$\therefore$ exact result = -[10's complement of 889]

$= -111$

iii $3570 - 2100$

Here,

$M = 3570$ $N = 2100$

~~Then~~ Now,

9's complement of $N = N' = 9999 - 2100$

$= 7899$

then

$M + N'$

$M = 3570$

$N' = +7899$

11469

$+1$

1470

$\therefore$ Answer = 1470

Again,

10's complement of $N = N' = 10000 - 2100$
$= 7900$

then,

$M + N'$

$M = 3570$
$N' = +7900$
$\overline{11470}$

$\therefore$ Answer = 1470

iv 20 - 100

Here

$M = 020 \qquad N = 100$

Now,

9's complement of $N = N' = 999 - 100$
$= 899$

then

$M + N'$

$M = 020$
$N = +899$
$\overline{919}$

$\therefore$ ~~Answer = 919~~ Since there is no end carry

Exact answer = - (9's complement of 919)

$\therefore$ Answer = - 80

Again,

10's complement of $N = N' = 1000 - 100$
$= 900$

then

$M + N'$

$M = 020$

$N' = +900$

920

since there is no end carry

exact answer = - (10's complement of 920)

$\therefore$ Answer = -80

Date
Page

5 Subtract the following binary numbers using 1's and 2's complement:

i 11010 - 1101

Here,

M = 11010

N = 01101

1's complement of N = N' = 11111 - 01101

= 10010

Then,

M = 11010 — 11010 + 1

N' = + 10010 — + 10010

11111 — 101100

+1

01101

∴ Answer = 01101

Again,

2's complement of N = N' = 10000 - 01101

= 10011

Then,

M = 11010

N' = + 10011

101101

Discarding the end carry

∴ Answer = 01101

ii 11010 - 10000

Here

M = 11010 N = 10000

Now,

1's complement of N = N' = 11111 - 10000
= 01111

Then

```
      1 1 1
M  =  1 1 0 1 0
N' = + 0 1 1 1 1
     -------------
     1 0 1 0 0 1
       └─ + 1
     -------------
     0 1 0 1 0
```

∴ Answer = 01010

Again,

2's complement of N = N' = 100000 - 10000
= 10000

Then:

```
M  =  1 1 0 1 0
N' = + 1 0 0 0 0
     -------------
   1) 0 1 0 1 0
```

Discarding the end carry

∴ Answer = 01010

iii 10010 - 1001)

Here,

M = 10010 N = 1001)

Now,

1's complement of N = N' = 11111 - 1001)
= 01100

Then,

M = 10010
N' = + 01100
11110

Since there is no end carry

exact answer = - of 1's complement of 11110

= - 00001

Again

2's complement of N = N' = 100000 - 1001)
= 01101

Then,

M = 10010
N' = + 01101
11111

Sinle there is no end carry

exact answer = - of 2's complement of 11111

= - 00001

iv 100 - 110000

Here

M = 000100 N = 110000

Now,

1's complement of N = N' = 111111 - 110000

= 001111

Then,

M = 000100 since there is no

N' = + 001111 end carry

010011

exact answer = - {1's comp of 010011}

= - 101100

Again,

2's complement of N = N' = 1000000 - 110000

= 010000

Then,

M = 000100

N' = + 010000

010100

Since there is no end carry

exact answer = - {2's complement of 010100}

= - 101100

Date
Page

6 Define analog ~~or~~ signal and digital signal. Write differences between analog and digital system.

→ Those signals which can be specified in every instant of time are called analog signals. and Those signals which can be specified in certain instant interval of time are called digital signals.

The difference between analog and digital systems are listed below.

	Analog System		Digital System
1	Any device which processes continious elements of information is called analog system	1	Any device which processes discrete element of information is called digital system
2	They are represented by any digits within a range	2	They are represented by binary digits 0 & 1.
3	They are usually represented by smooth wave's	3	They are usually represented by square waves.
4	Analog data is difficult to store	4	Digital data is easier to store.
5	for eg: Analog clock, Mercury		

	thermometer, magnetic tape reorder, etc.	5	for eg: Digital clock, Digital thermometer, CD etc

7 Define analog and digital systems. Why do you prefer digital systems over its analog counterparts?

→ An analog system is a system that processes continuous signals or data that can vary smoothly and take on any value within a given continuous range.

A digital system is a system that processes discrete signals or data represented in the form of binary digits (0 and 1) or distinct voltage levels.

I prefer digital systems over its analog counterparts because of the following reasons:

i Higher immunity to noise

→ Analog signals are continuous and easily distorted by environmental noise, static, or electrical interference, which degrades signal quality. whereas digital signals use binary digits (0 and 1) making them extremly robust against noise because small fluctuations do not affect the core binary value.

ii Superior Accuracy and precision

-> Analog systems are limited by the physical tolerances of their components and are prone to drift caused by temperature or aging. Digital systems can achieve virtually limitless precision simply by increasing the number of bits used to represent data.

iii Ease of Storage and Manipulation

-> Digital data can be easily stored on computer media (such as SSDs and flash memory) and manipulated, compressed or encrypted using software algorithms. Analog storage (like magnetic tapes) degrades over time and is harder to edit without loss of quality.

iv Programmability and Versatility

-> Digital systems are controlled by software and firmware, meaning the same hardware can perform entirely different tasks simply by changing the program. Analog systems require physical rewiring or hardware changes to alter their functions.

Date
Page

8 'Excess -3' code is also known as self-complementing code'.

→

Decimal	Excess 3
0	0011
1	0100
2	0101
3	0110
4	0111
5	1000
6	1001
7	1010
8	1011
9	1100

Here, 4 and 5 are 9's complement of each other their corresponding excess 3 code (0111 and 1000) are One's complement of each other. The same phenomenon can be seen for the pairs 3 and 6, 2 and 7, 1 and 8 and 0 and 9. Hence excess 3 code is also known as self complementing code.

Date
Page

9 Define: weighted and non-weighted codes with examples.

→ The codes which have positional value and each bit in the code has certain weightage is known as weighted code. for example: BCD, 8 4 -2 -1 code, 2421 code.

whereas

The codes which have no positional value and bits don't hold any weightage is known as Non-weighted code. for example Excess 3 code, Grey code and Alphanumeric code.

Date
Page

10 Show your familiarization with gray code. Write the 4-bit gray code. Also explain why Grey code is also known as reflect code.

→

0 0 0 0
0 0 0 1
0 0 1 1
0 0 1 0
0 1 1 0
0 1 1 1
0 1 0 1
0 1 0 0
1 1 0 0
1 1 0 1
1 1 1 1
1 1 1 0
1 0 1 0
1 0 1 1
1 0 0 1
1 0 0 0

4 bit gray code

Grey code is Known as reflect code because every larger code is created by reflecting the previous code in a mirror and changing the leading bit.

Date
Page

11 Perform the following

i $(89.9)_{10} = (\text{---})BCD, (\text{---})excess\ 3$

→

8	9	9
↓	↓	↓
1000	0011	1001

$\therefore (89.9)_{10} = (10000011.1001)_{BCD}$

8	9	9
↓	↓	↓
1011	0111	1100

$\therefore (89.9)_{10} = (1011\,0111.1100)_{excess\ 3}$

ii $(1101)_{gray} = (\text{---})_2$

→ take MSB

1 1 0 1
↓ ↓ ↓ ↓
1 0 0 1

$\therefore (1101)_{gray} = (1001)_2$

iii $(110101)_{Gray} = (---)_2$

Take MSB

1 1 0 1 0 1

1 0 0 1 1 0

$\therefore (110101)_{Gray} = (100110)_2$

iv $(101111)_2 = (---)_{Gray}$

Take MSB

1 0 1 1 1 1

1 1 1 0 0 0

$\therefore (101111)_2 = (111000)_{Gray}$

v $(63)_8 = (---)_{excess\ 3}$

$63 = 6 \times 8^1 + 3 \times 8^0$

$= (51)_{10}$

5 1

0101 0100

$\therefore (63)_8 = (10000100)_{excess\ 3}$

Date
Page

12 Write short notes on:

i Alphanumeric code

-> The binary code that is used to represent the 26 alphabets (A-Z), 10 numerals (0-9) & other special characters. (#, <, >, !) is known as alphanumeric code.

If we consider only 26 uppercase alphabets, 10 decimal digit & other special characters, we need 6 bits.

If we consider both upper & lower case alphabets, 10 decimal digits & other special characters, we need 7-bits

eg: ASCII → American standard code for Information Interchange.

ii Instruction code:

->

-> An instruction code is a group of bit that instructs a computer or a processor to perform a specific operation. Instruction code consists of operation codes also known as op-code and the operation part which specify the address of operand & result.

eg: MVI A, 20H